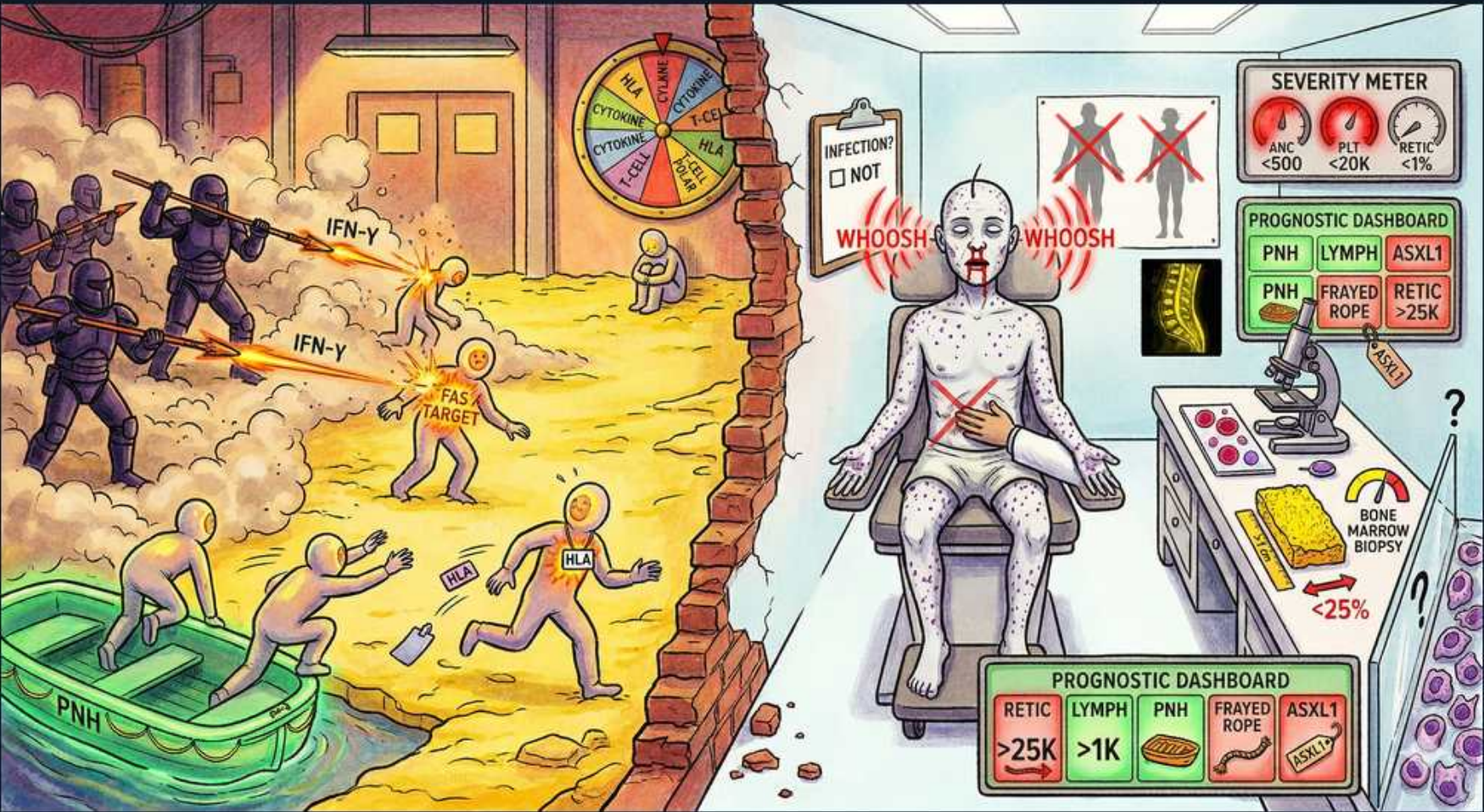


Bone Marrow Failure — The Immune Siege (AA Pathophysiology & Diagnosis)



1. T-cell soldiers

WHAT YOU SEE

Dark armored soldiers firing on the left

WHAT IT ENCODES

Central mechanism of acquired AA: autoreactive cytotoxic CD8+ T LYMPHOCYTES (oligoclonal, often expanded) recognize antigens on hematopoietic stem and progenitor cells and destroy them, emptying the marrow. This is an immune attack, not a primary stem-cell defect — the rationale for immunosuppressive therapy.

BOARD TRAP

Boards contrast this with a stem-cell-intrinsic disorder: acquired AA is predominantly IMMUNE-mediated (responds to ATG/cyclosporine). Inherited marrow failure (Fanconi, dyskeratosis congenita) IS an intrinsic stem-cell/DNA-repair/telomere defect — different management.

2. IFN-gamma fire

WHAT YOU SEE

Beams labeled IFN- γ aimed at marrow cells

WHAT IT ENCODES

Activated T cells secrete the inhibitory cytokines IFN- γ and TNF- α , which suppress proliferation of CD34+ progenitors, upregulate Fas on target cells, and trigger apoptosis — the molecular 'firepower' behind marrow destruction.

BOARD TRAP

IFN- γ /TNF- α -driven Th1 immunity is the lever immunosuppression pulls. This cytokine milieu also explains why eltrombopag (TPO-receptor agonist) can stimulate the surviving stem cells despite the suppressive environment.

3. Fas target

WHAT YOU SEE

Falling figure marked FAS TARGET

WHAT IT ENCODES

Stem/progenitor cells are killed largely by FAS–FAS-ligand–mediated APOPTOSIS, producing the profoundly hypocellular (fatty) marrow that defines aplastic anemia.

BOARD TRAP

The marrow is empty because of increased APOPTOSIS/immune killing, not because cells are maturing and being consumed peripherally — hence a LOW reticulocyte count (underproduction), the opposite of hemolysis.

4. HLA flags

WHAT YOU SEE

Fleeing figures carrying HLA tags

WHAT IT ENCODES

There are HLA associations (classically HLA-DR15/DR2), and immune-escape biology: clones that lose or alter HLA expression (6p copy-neutral loss of heterozygosity) evade the T-cell attack and can expand — one route to clonal hematopoiesis in AA.

BOARD TRAP

HLA-DR15 positivity and a PNH/HLA-loss clone predict better response to immunosuppression, but clonal escape (PNH, HLA-LOH, or MDS-type clones) also flags risk of later clonal EVOLUTION.

5. PNH escape boat

WHAT YOU SEE

Green boat labeled PNH carrying escapees

WHAT IT ENCODES

GPI-anchor-deficient PNH clones (PIGA-mutant, lacking CD55/CD59) 'escape' the immune attack because they lack the GPI-anchored target antigen, so they survive and expand within the failing marrow — explaining the strong AA–PNH overlap.

BOARD TRAP

Screen all AA patients with flow cytometry for a PNH clone at diagnosis and on follow-up. A PNH clone supports immune pathogenesis but also requires monitoring for hemolysis and thrombosis.

6. Pancytopenia man

WHAT YOU SEE

Pale seated patient with petechiae and WHOOSH nosebleeds

WHAT IT ENCODES

Clinical presentation = consequences of PANCYTOPENIA: anemia → fatigue, pallor, dyspnea; thrombocytopenia → petechiae, mucosal bleeding, epistaxis, easy bruising; neutropenia → infection risk. Onset is usually insidious; the marrow does not infiltrate, so there is typically NO hepatosplenomegaly or lymphadenopathy.

BOARD TRAP

Absence of organomegaly/lymphadenopathy helps separate AA from leukemia/lymphoma. Hepatosplenomegaly or blasts on smear argue AGAINST aplastic anemia and toward an infiltrative/clonal process.

7. Infection NO

WHAT YOU SEE

Clipboard reading INFECTION? NO

WHAT IT ENCODES

At presentation patients are often afebrile/without active infection despite leukopenia, but NEUTROPENIA (especially ANC <500) makes them highly vulnerable; febrile neutropenia is a medical emergency requiring urgent broad-spectrum antibiotics.

BOARD TRAP

Don't be reassured by an initially well-appearing patient — severe neutropenia means a fever must be treated as life-threatening sepsis immediately, even without a localizing source.

8. Empty marrow biopsy

WHAT YOU SEE

BONE MARROW BIOPSY pointing to a hypocellular fatty core, <25%

WHAT IT ENCODES

DIAGNOSIS requires peripheral PANCYTOPENIA plus a HYPOCELLULAR marrow on BIOPSY (cellularity <25%, fatty replacement) with NO fibrosis, NO infiltration, and NO dysplasia/blast excess. A core biopsy (not just aspirate) is essential to judge cellularity.

BOARD TRAP

The classic exam pitfall: AA marrow is HYPOcellular. A HYPERcellular or fibrotic/infiltrated marrow excludes AA and points to MDS, leukemia, myelofibrosis, or marrow infiltration.

9. Severity meter

WHAT YOU SEE

SEVERITY METER dials <500, <20K, <1%

WHAT IT ENCODES

CAMITTA severity criteria (need ≥2 of 3 in a hypocellular marrow): ANC <500/μL, platelets <20,000/μL, reticulocytes <20,000/μL (or <1%). SEVERE AA = meets criteria with ANC <500; VERY SEVERE AA = ANC <200.

BOARD TRAP

Severity (especially ANC <500 = severe, <200 = very severe) drives treatment urgency and choice: severe/very-severe disease in a young patient → urgent allo-HSCT; non-severe disease may be observed or treated with IST.

10. Reticulocyte low

WHAT YOU SEE

Dashboard RETIC >25K crossed with red low arrow

WHAT IT ENCODES

The reticulocyte count is LOW (corrected retic/RPI low; absolute retic <20–25,000), confirming marrow UNDERPRODUCTION rather than peripheral destruction or loss.

BOARD TRAP

A LOW reticulocyte count is the hallmark. A HIGH reticulocyte count would indicate hemolysis or blood loss (appropriate marrow response) and argues against aplastic anemia.

11. Exclude MDS (ASXL1)

WHAT YOU SEE

Prognostic dashboard chips PNH, LYMPH, ASXL1

WHAT IT ENCODES

Must EXCLUDE hypoplastic MDS — look for dysplasia, abnormal cytogenetics (e.g., monosomy 7, del(5q)), and somatic mutations. Mutations like ASXL1, DNMT3A, and RUNX1 suggest MDS-type clonal hematopoiesis and predict higher risk of progression, whereas BCOR/BCORL1 and PIGA mutations favor immune AA.

BOARD TRAP

Hypocellular MDS is the great mimic of AA. Cytogenetics + a myeloid mutation panel are needed; finding monosomy 7 or high-risk mutations (ASXL1/RUNX1) reclassifies the case as MDS and changes prognosis/therapy.

12. Frayed rope telomere

WHAT YOU SEE

FRAYED ROPE chip on the dashboard

WHAT IT ENCODES

SHORT/FRAYED TELOMERES flag an inherited telomere-biology disorder (dyskeratosis congenita; TERT/TERC mutations) presenting as 'apparently acquired' AA. Telomere length testing is part of the young-patient workup.

BOARD TRAP

Telomeropathies do NOT respond well to immunosuppression and have higher transplant toxicity (lung/liver) — missing them is a real management error. Look for the mucocutaneous triad (nail dystrophy, oral leukoplakia, reticular skin pigmentation) and pulmonary fibrosis.

13. Hairy cell exclusion

WHAT YOU SEE

Purple atypical lymphocytes in lower-right corner

WHAT IT ENCODES

Also exclude infiltrative/clonal causes of a deceptively 'empty-looking' marrow: HAIRY CELL LEUKEMIA (pancytopenia, monocytopenia, splenomegaly, dry tap from fibrosis), large granular lymphocytic leukemia, lymphoma, and hypoplastic leukemia.

BOARD TRAP

Hairy cell leukemia classically causes pancytopenia with MONOCYTOPENIA and splenomegaly and a fibrotic 'dry-tap' marrow — flow cytometry (CD11c/CD25/CD103, BRAF V600E) distinguishes it from true aplastic anemia.